

Bewley Banks

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Summary

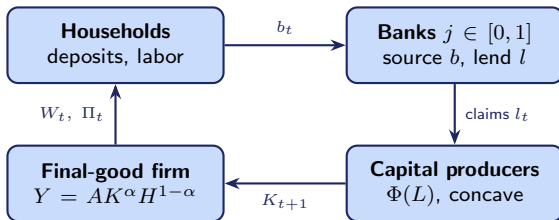
Central question

- How do movements in the **distribution of bank size and income** affect the macroeconomy?
- Existing macro-banking models use a **representative intermediary** — the cross-section of banks plays *no* role (Gertler–Kiyotaki 2010; Gertler–Karadi 2011).
- Heterogeneity leads to **concentration** in the banking sector and **amplification** of real and financial fluctuations

Methodology

- Introduce bank heterogeneity which drives fluctuations
- Connect Gertler-Kiyotaki *banking block* with Bewley–Aiyagari *incomplete markets*
- Introduce **Marginal Propensity to Lend (MPL)** as a *sufficient statistic* for aggregate dynamics
- Use Krussel-Smith approximation for the banking sector

Four agents



Real side = standard RBC, except:

- $A_t \in \{A_H, A_L\}$ – 2-state Markov TFP
- Capital producers use a **concave** $\Phi(\cdot)$, so the price of capital

$$Q_t = [\Phi'(L_t)]^{-1}$$

rises as aggregate lending L_t falls.

- This is the channel through which **bank distress hits the real economy**.

Households

Households

$$U(C_t, H_t) = \frac{1}{1-\psi} \left(C_t - \chi_1 \frac{H_t^{1+\chi_2}}{1+\chi_2} \right)^{1-\psi}$$

Subject to

$$C_t + \int_0^1 b_t(j) dj \leq \int_0^1 R_t b_{t-1}(j) dj + H_t W_t + \Pi_t + T_t$$

Deposit Euler equation pins down the **risk-free rate**:

$$R_{t+1} = \left[\mathbb{E}_t \beta u'(C_{t+1}) / u'(C_t) \right]^{-1}, \quad H_t = (W_t / \chi_1)^{1/\chi_2}.$$

Where:

ψ – risk aversion χ_1 – labor disutility $1/\chi_2$ – Frisch elasticity β – discount factor

Firms

Firms — competitive, identical Cobb–Douglas; produce the final good Y_t

$$Y_t = A_t K_t^\alpha H_t^{1-\alpha}, \quad 0 < \alpha < 1$$

There is a continuum of identical capital producing firms indexed by i , which rely on banks for financing. Bank level claims on firms $l_t(j)$ get aggregated into the stock of total claims: $L_t = \int_0^1 l_t(j) dj$. Each i takes $L_t(i)$ and produces $\Phi(L_t(i))$.

$$\max_{L_t(i)} Q_t \Phi(L_t(i)) - L_t(i), \quad Q_t = [\Phi'(L_t)]^{-1}$$

Return on capital taken as given by banks:

$$R_{t+1}^k = \frac{A_{t+1} \alpha K_{t+1}^{\alpha-1} H_{t+1}^{1-\alpha}}{Q_t}.$$

Banks - balance sheet

A continuum of banks j . Each starts a period with net worth $n_t(j)$.

Balance sheet identity

$$\underbrace{b_t(j)}_{\text{deposits}} + \underbrace{n_t(j)}_{\text{net worth}} = \underbrace{l_t(j)}_{\text{claims (loans)}}$$

Total portfolio return carries *both* kinds of heterogeneity:

$$R_t^T(j) = 1 + \kappa(j) \xi_t(j) r_t^k$$

- $\kappa(j) \sim \mathcal{P}(\alpha_\kappa, k_m)$ – *permanent* type, Pareto / Zipf ($\alpha_\kappa = 1$) — ex-ante
- $\xi_t(j)$ – *transitory* idiosyncratic shock, AR(1) Gaussian — ex-post

$$\xi_t(j) = (1 - \rho_\xi)\mu_\xi + \rho_\xi\xi_{t-1}(j) + \epsilon_t(j), \quad \epsilon \sim N(0, \sigma_\xi)$$

- r_t^k – common aggregate net return on capital

Banks - two crucial frictions

1. Convex non-interest cost

$$\text{cost} = \zeta_1 l_t(j)^{\zeta_2}, \quad \zeta_1 > 0, \zeta_2 > 1$$

Law of motion of net worth:

$$n_{t+1} = R_{t+1}^T l_t - R_{t+1} b_t - \zeta_1 l_t^{\zeta_2}$$

This **breaks scale invariance**. Now bank size (net worth) is a relevant state variable.

2. Agency friction (Gertler–Kiyotaki)

Banker can divert fraction λ of franchise value V_t . To deter diversion, assets are capped by franchise value:

$$\lambda l_t(j) \leq V_t(j)$$

$\Rightarrow$ an endogenous, occasionally-binding *leverage constraint*.

Banks are risk-neutral, survive with prob. σ , and pay out their franchise as dividends on exit.

Recursive bank problem

Aggregate state (Γ, A) . The key object:

Γ — the cross-sectional distribution of banks

A measure on $\mathcal{B} = N \times \Theta \times \Xi$, i.e. the joint distribution over **net worth** n , **permanent type** κ , **transitory shock** ξ .

Law of motion: $\Gamma' = F(\Gamma, A, A')$.

Bellman equation:

$$V(n, \kappa, \xi; \Gamma, A) = \max_{\{l, b, n'\} \geq 0} \beta \mathbb{E}[(1 - \sigma)n' + \sigma V(n', \kappa, \xi'; \Gamma', A') \mid \xi, A]$$

subject to

$$n' = [1 + \kappa \mathbb{E}(\xi' r^{k'})]l - R'b - \zeta_1 l^{\zeta_2}, \quad b + n = l, \quad \lambda l \leq V, \quad \Gamma' = F(\Gamma, A, A').$$

σ – survival probability $(1 - \sigma)$ – exit / dividend rule the franchise value enters its own constraint $\Rightarrow$ fixed point in V .

Partial-equilibrium lending policy — vs. Gertler–Kiyotaki

Substituting the constraints and letting the leverage constraint bind:

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$$L = \frac{\mathbb{E}\{\Omega(R'n - \zeta_1 l^{\zeta_2})\}}{\lambda - \mathbb{E}\{\Omega(1 + \kappa \xi' r^{k'} - R')\}}$$

Gertler–Kiyotaki

$$L^{GK} = \frac{\mathbb{E}\{\Omega R'\} n}{\lambda - \mathbb{E}\{\Omega(1 + r^{k'} - R')\}}$$

Two differences

- ① **Cost term** $\zeta_1 l^{\zeta_2}$ in the numerator $\Rightarrow$ scale matters, n is a state.
- ② κ, ξ in the denominator $\Rightarrow$ lending varies by *type* and *shock*.

Relevant idiosyncratic state vector: (n, κ, ξ) , vs. scalar n in GK.

$\Omega \equiv \beta(1 - \sigma + \sigma V(n', \kappa, \xi'; \cdot)/n')$: bankers' augmented discount factor.

Recursive competitive equilibrium

A RCE consists of a law of motion $\Gamma' = F(\Gamma, A, A')$, bank functions (V, L) , and household policies (C^*, H^*, B^*) such that, given prices $\{R, R^k, Q, W\}$:

- 1 value & policy functions solve each agent's problem;
- 2 Γ is **consistent** with agents' optimization;
- 3 prices satisfy the first-order conditions;
- 4 all markets clear.

Market clearing:

$$Y_t = C_t, \quad K_{t+1} = \Phi \left(\int L(n, \kappa, \xi) \Gamma_t dn d\kappa d\xi \right),$$

labor market clears by Walras' law.

The problem

To solve the bank's problem, each bank must **forecast** Γ' — an infinite-dimensional object.

Why is the high dimension of Γ the problem?

The chain of dependence

Bank chooses $l_t \Leftarrow$ needs $r_{t+1}^k \Leftarrow K_{t+1} \Leftarrow L_{t+1} = \int L d\Gamma_{t+1} \Leftarrow \Gamma_{t+1}$

- Under aggregate uncertainty, the future return on capital depends on what every bank does next period.
- That depends on the **entire future distribution** Γ_{t+1} .
- Γ is endogenous, time-varying, and **infinite-dimensional** - and a state variable

Curse of dimensionality

The Krusell–Smith solution: track one moment

Bounded-rationality assumption

Banks forecast using only the **first moment** $\bar{N}$ of the net-worth distribution.
Conjecture *log-linear* forecasting rules, conditional on the aggregate state:

$$\begin{aligned} A = A_L : \quad \log \bar{N}' &= \beta_{L,0}^N + \beta_{L,1}^N \bar{N}, & \log \bar{L}' &= \beta_{L,0}^L + \beta_{L,1}^L \bar{N} \\ A = A_H : \quad \log \bar{N}' &= \beta_{H,0}^N + \beta_{H,1}^N \bar{N}, & \log \bar{L}' &= \beta_{H,0}^L + \beta_{H,1}^L \bar{N} \end{aligned}$$

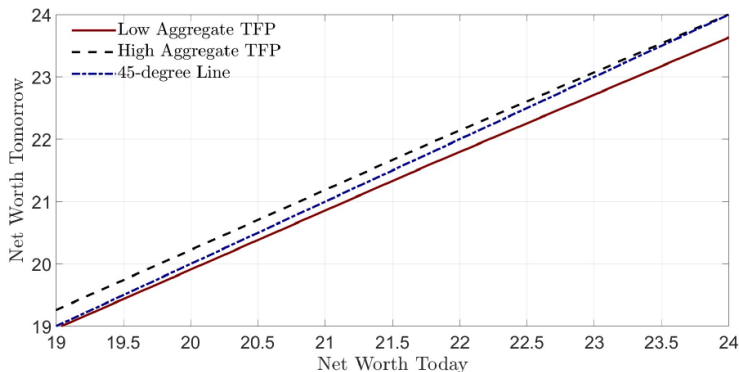
Algorithm (fixed point on β):

- 1 Guess $\beta \Rightarrow$ implied r^k path $\Rightarrow$ solve bank problem (VFI) for (V, L) .
- 2 Simulate a panel of 2,000 banks for 2,000 periods.
- 3 Re-run OLS of $\log \bar{N}_t, \log \bar{L}_t$ on $\log \bar{N}_{t-1} \Rightarrow$ update β .
- 4 Iterate to convergence.

Self-consistency: the rule banks *use* must equal the rule their behavior *generates*.

Why one moment is enough - $R^2 > 0.999$

Figure 2: Tomorrow's vs Today's Aggregate Net Worth



Notes: Tomorrow's vs. today's aggregate bank net worth for the baseline model. The top (bottom) line corresponds to the law of motion of aggregate net worth in a high (low) TFP state. The middle line is the 45-degree line.

The Marginal Propensity to Lend (MPL)

Definition

MPL = change in bank-level lending from a marginal change in own net worth, $\partial l / \partial n$.

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Gertler–Kiyotaki

$$\text{MPL} = \frac{\mathbb{E}\{\Omega R'\}}{\lambda - \mathbb{E}\{\Omega(1 + \kappa \xi' r^{k'} - R')\} + \zeta_1 \zeta_2 l^{\zeta_2 - 1}} \quad \text{MPL}^{GK} = \frac{\mathbb{E}\{\Omega R'\}}{\lambda - \mathbb{E}\{\Omega(1 + r^{k'} - R')\}}$$

- In GK, MPL is a **single aggregate number** — no cross-sectional variation.
- Here, MPL is **increasing in κ and ξ** , and **decreasing in size l** .
- Intuition: small, constrained banks lend more aggressively out of an extra dollar of net worth.

The average MPL is the sufficient statistic for aggregate amplification.

Calibration

- One period = a quarter.
- Banking moments from U.S. **Call Reports**; aggregates from FRED.
- κ Pareto with $\alpha_\kappa = 1$ **Zipf's law** in the data
- Re-calibrated separately for every model variant to hit the *same* targets.
- IRFs via "MIT shock" - one-SD TFP innovation
- Crises via 10000-period simulation - crisis defined as output falling below a threshold

Representative vs Heterogeneous banks

Figure 5: Marginal Propensity to Lend in the Distribution

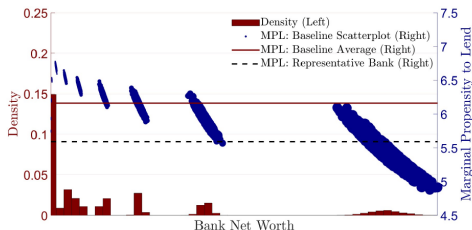


Figure 5 — net-worth density (left axis) & MPL across banks (right axis).

Only ex-ante heterogeneity changes the result.

(a) “As-if” — ex-post only

- With only ξ shocks, the distribution moves but stays **near-symmetric**.
- Net worth accumulates among banks with *the same lending slope*.
- Average MPL $\approx$ representative-bank MPL $\Rightarrow$ **exact aggregation**.

(b) Amplification — add ex-ante

- Pareto κ creates a fat right tail.
- Few huge low-MPL banks, but *many* small high-MPL banks.
- The latter dominate $\Rightarrow$ **average MPL > rep-bank MPL** $\Rightarrow$ shocks amplified.

MPL in two special cases

Figure 4: Marginal Propensity to Lend

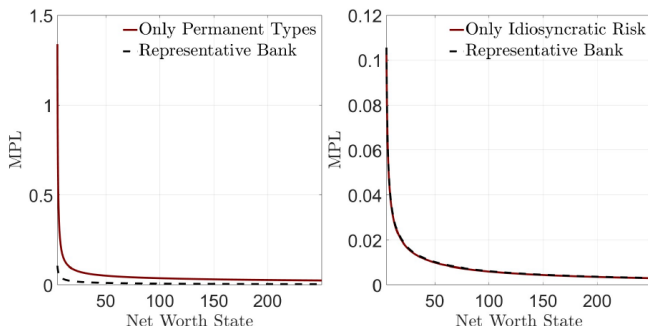
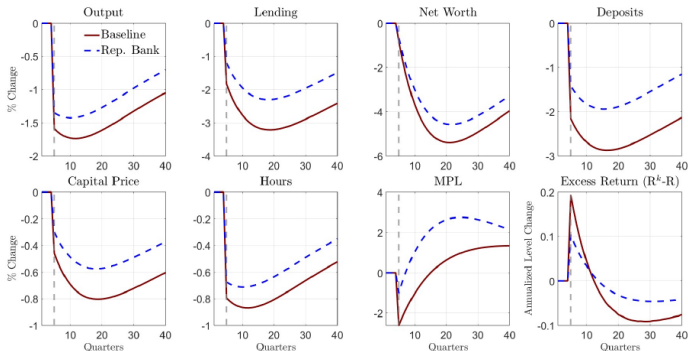


Figure 4 — MPL vs. net worth. *Left*: only permanent κ (MPL $>$ rep-bank). *Right*: only idiosyncratic ξ (MPL $\approx$ rep-bank: the as-if result).

Result 1 - IRFs to a negative TFP shock

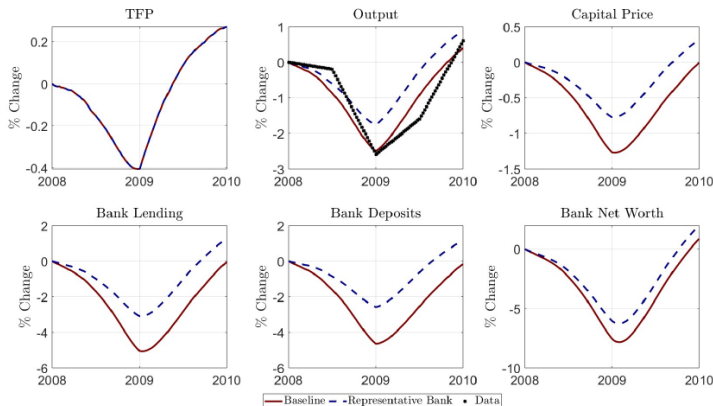
Figure 8: Impulse Responses to an Aggregate TFP Shock



Baseline (solid) vs. **representative bank** (dashed). Output contracts $\approx 20\%$ more under heterogeneity. Credit spread ($R^k - R$) spikes, franchise value falls, banking sector shrinks $\Rightarrow$ real recession.

Result 2 - simulating the Great Financial Crisis

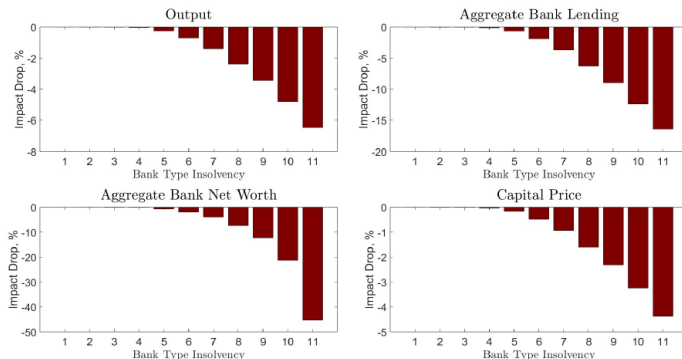
Figure 11: Simulating the Great Financial Crisis



Notes: Event study analysis of financial crisis events based on a long simulation of the baseline economy and of the representative-bank benchmark. An event is defined as a decline in aggregate output triggered by an aggregate Total Factor Productivity (TFP) shock falling below a specified threshold. The threshold is calibrated to match the 2009 empirical trough of output in the baseline model.

Granular bank failures (Too Big To Fail)

Figure 10: Granular Bank Failures



Notes: The aggregate impact of failures of banks belonging to a specific permanent return type κ . Failure corresponds to a shock that

Figure 10 — aggregate impact of the failure of each permanent type κ . Failure = a shock wiping out all net worth. Higher types $\Rightarrow$ much larger losses; small banks $\approx$ negligible.

Takeaways

Methodology

- Heterogeneity in bank types
- Convex cost makes net worth a state variable
- Krussel-Smith algorithm allows to track Γ
- Average MPL is the sufficient statistic - ex-post heterogeneity leads to as-if result, ex-ante leads to amplification

Results

- IRF - Output contracts by around 20% under heterogeneity
- The representative-bank economy **misses the 2009 output drop by $\approx \frac{1}{3}$.**
- Granular bank failures. Failure of the top type $\Rightarrow$ output -7% , lending -17% , net worth -45%